



KIC
ROBO-INVENT 2026

**PARTICIPANT DELEGATE
BOOKLET**



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WELCOME MESSAGE

Welcome to KIC Robo-Invent2026 – the first inter-school robotics championship hosted by the Kandy Innovation Center, NIBM.

This event was built around one simple idea: every student in the Kandy zone deserves a real chance to try robotics, regardless of whether their school already has a robotics club, a lab, or prior experience. That's why the bootcamp comes before the competition. to make sure every team starts from the same place, with the same foundation, and the same shot at doing well.

Over the next few months you'll learn to wire sensors, write logic that makes real-time decisions, and build a robot that can think for itself on a course it's never seen before. You'll hit problems you didn't expect - a sensor that misreads a color, a motor that doesn't respond the way your code assumes it should, a robot that gets stuck where you swore it wouldn't. That's normal. Every team in this competition will run into something like it.

What matters is what you do next. You'll debug it, test it again, and get it working and that process, not the trophy at the end, is the actual point of this event.

Robotics rewards patience and curiosity far more than it rewards getting things right on the first try. If you walk away from Robo-Invent 2026 having learned how to take a problem apart and put a working solution back together, you've already gotten the most valuable thing this event has to offer.

We're glad you're here. Read this guidebook carefully, ask questions early, lean on your bootcamp mentors when you're stuck, and enjoy building something that actually works.

ABOUT KANDY INNOVATION CENTRE

The Kandy Innovation Center (KIC) is the second center for creative technologies established in Sri Lanka, following the National Innovation Centre (NIC) in Kirulapona. KIC was founded through a strategic partnership between the National Institute of Business Management (NIBM), Limkokwing University of Creative Technology, and Coventry University, United Kingdom, strengthening NIBM's position as an international institution and connecting Sri Lankan students directly to the global education and job market.



The Kandy Innovation Center (KIC) NIBM has always been a pioneer in driving digital education and creative technologies in Sri Lanka. To foster computational thinking, practical problem-solving, and a sustainable passion for STEM (Science, Technology, Engineering, and Math) among school children, KIC proposes to host the inaugural "Kandy Robo-Invent 2026" championship. This inter-school robotics competition targets students across the Kandy district, offering them an introductory yet deeply engaging competitive platform to explore automation, logic design, and basic robotics.

The Media Club covers photography, videography, and content production for KIC's events and communications, giving students real production experience outside a classroom setting. The Chess Club gives students a space to sharpen strategic thinking and problem-solving in a more traditional setting, complementing the technical and creative work happening elsewhere on campus.

ABOUT ROBO - INVENT 2026

Kandy Robo-Invent2026 is the inaugural inter-school robotics championship hosted by the Kandy Innovation Center (KIC), part of the National Institute of Business Management (NIBM). Open to Ordinary Level (O/L) & Advanced Level (A/L) students across the Kandy education zone. The competition gives students a hands-on, accessible introduction to automation, logic design, and robotics, regardless of whether their school already has a robotics program. Each participating school may register up to two teams, with a maximum of three students per team. To ensure equal opportunities for all participants, schools are not required to have prior robotics experience or an established robotics club. However, robotics kits and other competition equipment will not be provided, and participating teams will be responsible for arranging their own required hardware and materials for the competition.

OUR OBJECTIVES

PROMOTE STEM EDUCATION

- Give students real, hands-on practice with logical reasoning, basic electronics, and algorithmic thinking, not just theory from a textbook.

SKILL DEVELOPMENT

- Build foundational skills in automation, sensor integration, and programming, using both block-based and text-based environments, so students leave with abilities they can keep building on.

A STRONGER KANDY TECH COMMUNITY

- Help establish KIC NIBM as a regional hub for computing, creative design, and technology education in the Central Province, creating more opportunities like this one for students in the years ahead.

A PATH BEYOND O/L

- Introduce students to what's possible after O/L through hands-on exposure to KIC's programs in computing and technology, for those who want to keep going in this direction.



THE CHALLENGE

The Smart Path Navigation Challenge is designed to evaluate a robot's autonomous navigation, decision-making, and obstacle avoidance capabilities in a dynamic environment. The competition arena consists of a grid of equally sized squares, similar to a chessboard, where each square represents a possible position for the robot.

At the beginning of the challenge, the robot is placed on the designated START square. The arena contains Green markers, which indicate the valid route to the destination, and Red markers, which represent restricted areas that the robot must never enter. In addition, one or more obstacles will be placed at random positions along the route, requiring the robot to detect and safely navigate around them while continuing toward its destination.

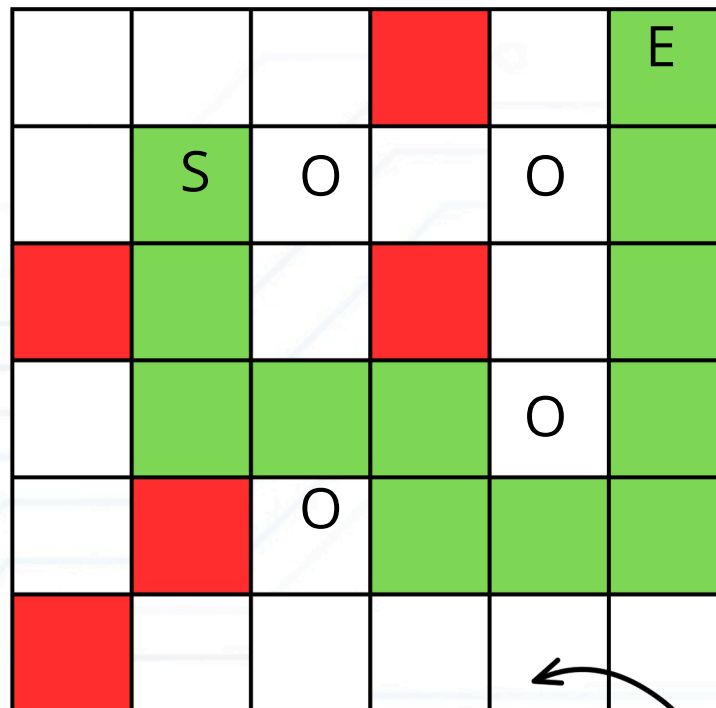
The robot must autonomously identify the correct path by recognizing the Green markers and make real-time navigation decisions throughout the mission. It is not permitted to follow or move onto any Red-marked square. If an obstacle blocks the intended route, the robot must detect its presence, determine an alternative safe path, and continue its journey without human intervention.

The mission is successfully completed when the robot reaches the designated FINISH square while adhering to all competition rules. Teams are encouraged to optimize both the accuracy and efficiency of their navigation algorithms, as the overall completion time will be a key factor in the competition.

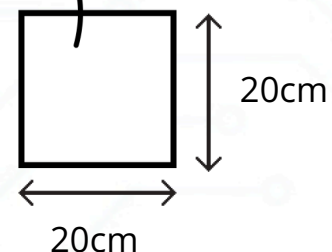


ARENA SPECIFICATIONS

ARENA SIZE: 120 CM × 120 CM



S = START
E = END
G = GREEN PATH
R = RED SQUARE
O = OBSTACLES



MISSION REQUIREMENTS

1. Start from the designated START square.
2. Detect and follow the route indicated by the Green markers.
3. Avoid entering any Red square at any point during navigation.
4. Detect and safely avoid obstacles encountered along the path.
5. Continue navigating autonomously until it reaches the FINISH square.
6. Complete the mission in the shortest possible time while following all competition rules



EXECUTION RULES

Each participating team will have to bring their own standard Robot Chassis Kit and the relevant components to ensure a fair and uniform competition.

1. The arrangement of Green and Red markers will be changed before each competition round.
2. The number and placement of obstacles may vary between rounds. Participating teams will be informed of the competition rules in advance but will not know the exact path configuration before the challenge begins
3. The robot must rely entirely on its onboard sensors and programmed algorithms to make navigation decisions during execution.
4. Hard-coded paths or predefined movement sequences are not permitted, as the arena configuration will change for every round.
5. The robot must demonstrate autonomous path planning, obstacle avoidance, and adaptive decision-making throughout the challenge.
6. The use of remote control is strictly prohibited once the competition has started.
7. Bluetooth or Wi-Fi communication for controlling the robot during the competition is not allowed.
8. No human intervention is permitted after the robot has been placed on the START square and the challenge has begun.
9. Any violation of the above conditions may result in penalty points or disqualification at the discretion of the judges
10. size limits to the robot chassis
 - Maximum length: 15 cm
 - Maximum width: 15 cm
 - Maximum height: 10 cm



SCORE CARD

HOW POINTS ARE AWARDED (OUT OF 100)



EVENT DAY SCHEDULE



INTRODUCTION TO THE CHALLENGE & RULES OVERVIEW

First meeting of all teams. Explaining the competition format, detailed team rules, and presenting the complete challenge scope.

BUILD & TEST

Teams build, program, and fine-tune their robots ahead of the finale.



GRAND FINALE

Live competition at KIC NIBM. Time trials, panel evaluation, and awards ceremony.



TEAM CONDUCT & SAFETY

CODE OF CONDUCT

Treat fellow competitors, mentors, and judges with respect. Unsportsmanlike conduct or interference with another team's robot may result in penalties or disqualification.

ELECTRICAL & BUILD SAFETY

Handle batteries, wiring, and soldering tools only under supervision. If a robot malfunctions in a way that could be unsafe, step away and alert event staff immediately.

CONTACT & SUPPORT

Faculty / staff advisor

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**GOOD LUCK”
ROBO-INVENT 2026
TEAMS BUILD BOLDLY..!**



**KIC
ROBO-INVENT 2026**

NIBM | THE CITY UNIVERSITY

KANDY
innovation
Centre